ESE 1205 Concept Primer

Thermodynamics for Energy Engineering · KUET · read this first: the subject from zero, before the past-paper solutions

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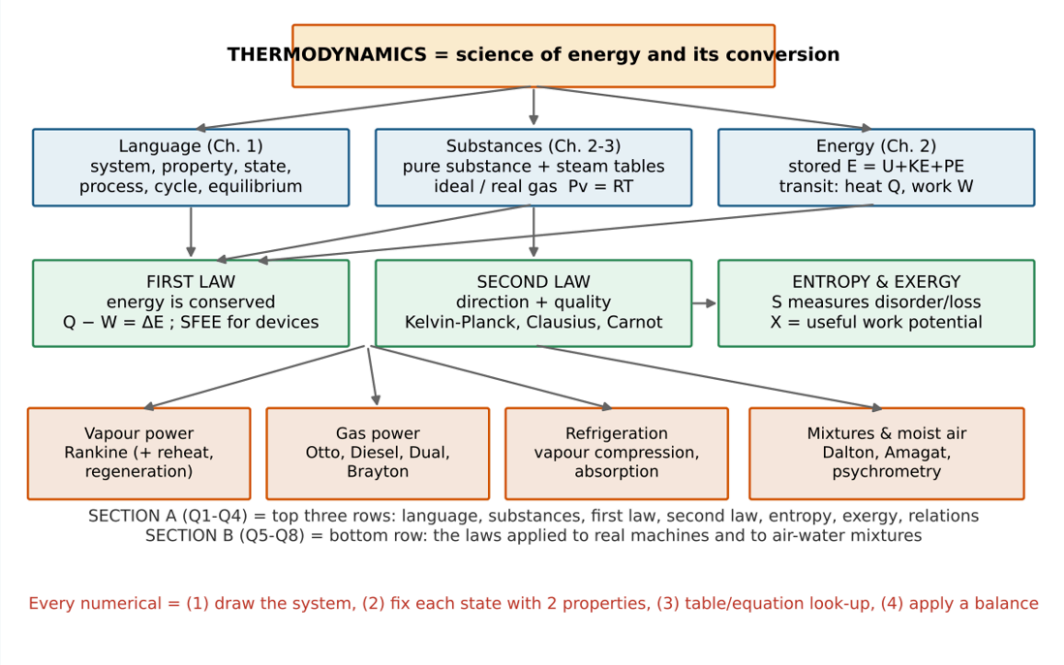
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How to use this primer

This booklet teaches thermodynamics from zero, in the order the exam needs it. Read it once before the Complete Solutions. Each section has a green **core idea** box (learn this first), the key terms, the formulas with every symbol and unit, and a short worked example. Section 2 is a maths and units refresher: do not skip it, because most lost marks in this course come from unit slips and table reading.

The pass set (from the Question Map): Section A topics A1 (basic concepts), A2 (energy, heat and work), A3 (pure substance), A5 (first law), A6 (device numericals), A7 (second law and Carnot), A8 (entropy); Section B topics B2 and B3 (Rankine theory and numericals), B4 (IC engine cycles), B7 (gas mixtures) and B8 (psychrometry). Those topics are enough to write three full questions in each section.

The whole course on one page



Concept map of ESE 1205: from basic definitions, through the laws, to the cycles and applications

Read the map from top to bottom: first you learn the **words** (system, property, state), then the **substances** (water/steam from tables, ideal gases from Pv = RT), then the **laws** (first law = energy is conserved, second law = direction and quality), and finally you apply them to **cycles** (steam plant, engines, gas turbine, refrigerator) and to **mixtures** (gas mixtures, moist air).

Maths, physics and units refresher

> Almost every numerical is: **write the right equation → get properties from a table or a formula → substitute with consistent units**. Use SI: kPa, m^3, kg, K, kJ. Then kPa × m^3 = kJ automatically.

Units you must convert without thinking

Quantity	Conversions
Pressure	1 bar = 100 kPa = 0.1 MPa; 1 atm = 101.325 kPa = 760 mm Hg; 1 MPa = 1000 kPa
Temperature	T(K) = t(°C) + 273.15. Use K in every gas law, efficiency and entropy formula. A temperature *difference* is the same in K and °C.
Energy, power	1 kJ = 1000 J; 1 kW = 1 kJ/s; 1 kWh = 3600 kJ
Volume	1 m^3 = 1000 L
Kinetic energy	$V^2/2$ in m^2/s^2 = J/kg; divide by 1000 for kJ/kg (100 m/s → 5 kJ/kg)
Potential energy	gz in J/kg; divide by 1000 (10 m → 0.098 kJ/kg)
Mass flow	$\dot{m} = \rho AV = AV/v$ (kg/s); per minute values: divide by 60

Logs and powers (needed for gas processes)

- $\ln(a/b) = \ln a - \ln b$; $\ln(a^n) = n \ln a$; $e^{\ln x} = x$. Example: $\ln 2 = 0.693$, $\ln 10 = 2.303$.
- Powers: $x^{0.2857}$ appears in every isentropic air problem because $(\gamma - 1)/\gamma = 0.4/1.4 = 0.2857$. On a calculator: 8 [$x^y$] 0.2857 = 1.811.
- To undo a power: if $r^{0.4} = 1.886$ then $r = 1.886^{1/0.4} = 1.886^{2.5}$.

Linear interpolation (needed for steam tables)

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y = y1 + (x - x1)(y2 - y1)/(x2 - x1)
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Example: h at 3 MPa, 355 °C, with h(350 °C) = 3115.3 and h(400 °C) = 3231.7: h = 3115.3 + (5/50)(116.4) = 3127.0 kJ/kg.

Reading the steam tables

1. Decide the region: compare the given T with T_{sat} at the given p (or v with v_f and v_g).
2. $T < T_{sat}$: compressed liquid → use $h \approx h_f$ at T. $T = T_{sat}$ with a quality: wet mixture → saturated table, $h = h_f + x\, h_{fg}$. $T > T_{sat}$: superheated table at p and T.
3. Saturated tables come in two forms: by temperature (enter with T) and by pressure (enter with p). Subscripts: f = saturated liquid, g = saturated vapour, fg = g − f.
4. For an isentropic process, carry s across: $s_2 = s_1$, then $x_2 = (s_2 - s_f)/s_{fg}$.

Calculus you need

Work is the area under a curve: $W = \int p\, dV$. For $p = C/V$ (isothermal ideal gas) $\int dV/V = \ln(V_2/V_1)$. For $p = C/V^n$, $\int V^{-n} dV = (V_2^{1-n} - V_1^{1-n})/(1 - n)$. A partial derivative $(\partial p/\partial T)_v$ means "how p changes with T while v is held constant".

Systems, properties and states (A1)

> Thermodynamics studies energy and its conversion. First choose the **system** (what you study) and its **boundary**. A system is **closed** if no mass crosses the boundary, **open** if mass flows through, **isolated** if neither mass nor energy crosses. A **property** (p, T, v, u, h, s) describes the state; a **process** changes the state; a **cycle** returns to the start.

- **Intensive** properties do not depend on size (T, p, ρ); **extensive** ones do (m, V, U). Small letters (v, u, h, s) are per kg.
- **Equilibrium**: nothing changes if the system is left alone; needs mechanical, thermal, chemical and phase equilibrium.
- **Quasi-equilibrium process**: so slow that the system passes through equilibrium states; it can be drawn as a line on a diagram.
- **Steady flow**: conditions inside a device do not change with time (a running turbine). **Transient**: they do (a tank being filled).

Symbol	Meaning	Unit
p	pressure (force per area)	kPa
T	absolute temperature	K
V, v	volume, specific volume V/m	m^3, m^3/kg
ρ	density = 1/v	kg/m^3
m, $\dot{m}$	mass, mass flow rate	kg, kg/s

Energy, heat and work (A2)

A system **stores** energy as kinetic (motion), potential (height) and internal energy U (molecular motion and bonds). Energy **crosses** the boundary only as **heat Q** (driven by a temperature difference) or **work W** (any other cause: moving piston, shaft, electricity). Q and W are not properties: their amounts depend on the path.

$$E = KE\ \text{(kinetic energy)} + PE\ \text{(potential energy)} + U = mV^2/2 + mgz + U$$

Symbol	Meaning	Unit
Q	heat transfer, + into the system	kJ
W	work, + done by the system	kJ
U, u	internal energy, per kg	kJ, kJ/kg
V	velocity (in KE)	m/s
z	height above a reference	m
g	9.81	m/s^2

Example: an electric heater in a room. If the system is the room air, energy enters as heat (the hot element warms the air). If the system includes the heater, energy enters as electrical work through the wire. Same energy, different label: always define the system first.

Pure substances and the steam tables (A3)

A **pure substance** has the same chemical composition everywhere (water, even as ice + water + steam). When water is heated at constant pressure it goes: compressed liquid → saturated liquid → wet mixture (boiling at constant T_{sat}) → saturated vapour → superheated vapour. The **T-v diagram** shows these regions under and around a dome. Two independent properties fix the state (state postulate).

$$x = m_{vapour}/m_{total} \qquad v = v_f + x(v_g - v_f) \qquad h = h_f + x\ h_{fg}$$

Symbol	Meaning	Unit
x	quality (dryness fraction), 0 to 1	–
T_{sat}, p_{sat}	boiling temperature at p, boiling pressure at T	°C, kPa
v_f, v_g	specific volume of saturated liquid, vapour	m^3/kg
h_{fg}	latent heat of vaporisation	kJ/kg
F, C, π	degrees of freedom, components, phases (F = C − π + 2)	–

Example: 1 kg of water at 100 kPa with x = 0.5: v = 0.001043 + 0.5(1.694 − 0.001043) = 0.848 m^3/kg.

Ideal and real gases (A4)

At low pressure and high temperature, gases obey **Pv = RT**. Each gas has its own R = R_u/M. Near the critical point or at high pressure, real gases deviate; the **compressibility factor Z = Pv/RT** measures this, and equations such as van der Waals and Redlich-Kwong correct for molecular attraction (a) and size (b).

$$PV = mRT = nR_uT \qquad R = R_u/M \qquad c_p - c_v = R \qquad \gamma = c_p/c_v$$

Symbol	Meaning	Unit
R_u	universal gas constant 8.314	kJ/kmol K
M	molar mass (air 28.97, N_2 28, O_2 32, CO_2 44)	kg/kmol
R	gas constant (air 0.287)	kJ/kg K
n	number of moles = m/M	kmol
c_p, c_v	specific heats (air 1.005, 0.718)	kJ/kg K
P_{cr}, T_{cr}	critical pressure, temperature	kPa, K

Example: 1 kg of air at 100 kPa, 300 K: V = 1 × 0.287 × 300/100 = 0.861 m^3.

The first law (A5) and devices (A6)

Energy is conserved. **Closed system:** Q − W = ΔU (+ ΔKE + ΔPE). **Steady-flow device:** $\dot{Q} - \dot{W} = \dot{m}[(h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(z_2 - z_1)]$. Enthalpy h = u + pv includes the flow work needed to push fluid in and out.

Symbol	Meaning	Unit
h	enthalpy per kg = u + pv	kJ/kg
$\dot{Q}$, $\dot{W}$	heat and power rates	kW
q, w	heat and work per kg of fluid = $\dot{Q}/\dot{m}$, $\dot{W}/\dot{m}$	kJ/kg
n	polytropic index in $pV^n = C$	–

Device	What it does	Energy equation (usual simplifications)
Turbine	fluid expands, turns a shaft	$w = h_1 - h_2$ (adiabatic)
Compressor / pump	raises pressure using work	$w_{in} = h_2 - h_1$; pump: $v(p_2 - p_1)$
Nozzle / diffuser	changes velocity	$h_1 + V_1^2/2 = h_2 + V_2^2/2$
Throttle valve	drops pressure	$h_1 = h_2$
Heat exchanger	heat passes from hot to cold stream	$\dot{m}_{hot}(h_{in} - h_{out}) = \dot{m}_{cold}(h_{out} - h_{in})$

Example: steam enters an adiabatic turbine with h_1 = 3248 kJ/kg and leaves with h_2 = 2361 kJ/kg: w = 887 kJ/kg; for 5 MW, $\dot{m}$ = 5000/887 = 5.6 kg/s.

The second law and heat engines (A7)

Heat flows naturally from hot to cold, and no engine can turn all the heat it receives into work: some must be rejected to a cold sink. The best possible (reversible, **Carnot**) engine between T_H and T_L has efficiency $1 - T_L/T_H$. All real engines are less efficient because of irreversibilities (friction, heat transfer across finite ΔT, mixing).

$\eta_{th} = W_{net}/Q_H = 1 - Q_L/Q_H$ $\eta_{Carnot} = 1 - T_L/T_H$ COP_R (coefficient of performance) $= Q_L/W_{in}$

Symbol	Meaning	Unit
Q_H, Q_L	heat from the source, heat to the sink	kJ
T_H, T_L	source and sink temperatures (absolute)	K
η	thermal efficiency	– or %

Example: a Carnot engine between 600 K and 300 K: η = 1 − 300/600 = 50 %; from 1000 kJ of heat it gives at most 500 kJ of work.

Entropy (A8)

Entropy S is a property that measures disorder; it increases when heat is added and whenever there is irreversibility. In an isolated system entropy can never decrease (ΔS ≥ 0). An ideal adiabatic process has constant entropy (**isentropic**), and real machines are compared with it by the isentropic efficiency.

$dS = (\delta Q/T)_{rev}$ $\quad T\,ds = du + p\,dv = dh - v\,dp$ $\quad S_2 - S_1 = \int \delta Q/T + S_{gen}$

Symbol	Meaning	Unit
S, s	entropy, per kg	kJ/K, kJ/kg K
S_{gen}, σ	entropy generated by irreversibility (≥ 0)	kJ/K
η_T, η_C	isentropic efficiency of turbine, compressor	–

Example: air compressed isentropically from 285 K with pressure ratio 8: $T_{2s} = 285 \times 8^{0.2857} = 516$ K. With $\eta_C = 0.8$ the real exit is 285 + 231/0.8 = 574 K.

Exergy and property relations (A9, A10)

Exergy is the maximum useful work a system can give as it comes into equilibrium with the environment (the **dead state**, 25 °C and 1 atm). Energy is always conserved, but exergy is destroyed by every irreversibility: $X_{destroyed} = T_0 S_{gen}$. **Maxwell relations** let us compute entropy changes from measurable p, v, T data.

Symbol	Meaning	Unit
X, φ	exergy of a closed system, per kg	kJ, kJ/kg
ψ	flow exergy per kg	kJ/kg
T_0, p_0	environment (dead-state) temperature and pressure	K, kPa
η_{II}	second-law efficiency = W_{useful}/W_{rev}	–

Power cycles: Rankine, Otto, Diesel, Brayton (B1-B5)

A **power cycle** repeats four processes: compress → add heat → expand → reject heat. The **Rankine cycle** (steam plant) pumps liquid, boils it, expands steam in a turbine and condenses it. The **Otto** (petrol), **Diesel** and **Dual** cycles model piston engines; the **Brayton** cycle models gas turbines. Efficiency rises when heat is added at a higher average temperature or rejected at a lower one.

Cycle	Heat added at	Efficiency
Rankine	constant p (boiler)	$[(h_3 - h_4) - w_p]/(h_3 - h_2)$
Otto	constant v	$1 - 1/r^{\gamma-1}$
Diesel	constant p	$1 - (1/r^{\gamma-1})(r_c^{\gamma} - 1)/[\gamma(r_c - 1)]$
Brayton	constant p	$1 - 1/r_p^{(\gamma-1)/\gamma}$

Symbol	Meaning	Unit
r	compression ratio V_{max}/V_{min}	–
r_c	cut-off ratio v_3/v_2 (Diesel)	–
r_p	pressure ratio p_2/p_1 (Brayton) or p_3/p_2 (Dual)	–
`MEP (mean effective pressure)`	mean effective pressure = $w_{net}/(v_1 - v_2)$	kPa
y	fraction of steam bled to a feedwater heater	–

Example: Otto cycle with r = 8: $\eta = 1 - 1/8^{0.4} = 1 - 1/2.297 = 56.5\ \%$.

Refrigeration (B6)

A refrigerator uses work to move heat from a cold space to warm surroundings. In **vapour compression**, a refrigerant evaporates (absorbing heat) at low pressure, is compressed, condenses (rejecting heat) at high pressure and is throttled back. In **vapour absorption**, the compressor is replaced by an absorber, pump and generator driven by heat.

Symbol	Meaning	Unit
`COP`	coefficient of performance = q_L/w_{in}	–
`TR (ton of refrigeration (3.5 kW))`	ton of refrigeration = 3.517 kW	kW
$h_1 \dots h_4$	enthalpies at evaporator exit, compressor exit, condenser exit, valve exit	kJ/kg

Gas mixtures and moist air (B7, B8)

A mixture of ideal gases behaves like one ideal gas with an average molar mass $M_m = \Sigma y_i M_i$. Each gas exerts its own **partial pressure** $p_i = y_i p$ (Dalton). **Moist air** is dry air + water vapour; its moisture is described by specific humidity ω, relative humidity φ and dew point, found from dry- and wet-bulb readings with Carrier's equation.

Symbol	Meaning	Unit
y_i	mole fraction n_i/n	–
mf_i	mass fraction m_i/m	–
p_v, p_s	vapour pressure, saturation pressure at DBT (dry-bulb temperature)	bar
ω	specific humidity = $0.622 p_v/(p - p_v)$	kg/kg dry air
φ	relative humidity = p_v/p_s	%
μ	degree of saturation = ω/ω_s	–
t_{db}, t_{wb}	dry-bulb, wet-bulb temperature	°C

Example: p_V = 0.02 bar at p = 1.0132 bar: ω = 0.622 × 0.02/0.9932 = 0.0125 kg/kg dry air (12.5 g of water per kg of air).

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
BDC	bottom dead centre
CI	compression-ignition
CM	control mass
COP	coefficient of performance
CV	control volume
DBT	dry-bulb temperature
DPT	dew-point temperature
EOS	equation of state
FWH	feedwater heater
HP	high-pressure
IAPWS	International Association for the Properties of Water and Steam
IC	internal combustion
KE	kinetic energy
kmol	kilomole (molar mass in kilograms)
KP	Kelvin-Planck (statement)
kPa	kilopascal
kW	kilowatt
LP	low-pressure
MEP	mean effective pressure
MPa	megapascal
MW	megawatt
PE	potential energy
PMM1	perpetual-motion machine of the first kind
PMM2	perpetual-motion machine of the second kind
RH	relative humidity
RK	Redlich-Kwong
SFEE	steady-flow energy equation
SI	spark-ignition
SRK	Soave-Redlich-Kwong
TDC	top dead centre
Tk	Bangladeshi taka
TR	ton of refrigeration (3.5 kW)
VAR	vapour absorption refrigeration
VCR	vapour compression refrigeration
vdW	van der Waals
WBT	wet-bulb temperature